THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. How many grams of salt are present in 250 g of a 4 per cent salt solution? 1

(a) 10 g **(b)** 40 g **(c)** 25 g **(d)** 4 g

2. Ammonia contains 82.35 per cent nitrogen. In 34 g of ammonia the mass of nitrogen is: 1

(a) 6 g **(b)** 28 g **(c)** 14 g **(d)** 17 g

3. An element has atomic number 19. The number of valence electrons is: 1

(a) 1 **(b)** 8 **(c)** 9 **(d)** 2

4. A body starts from rest with acceleration 2 m/s squared. The distance covered in 10 s is: 1

(a) 100 m **(b)** 10 m **(c)** 200 m **(d)** 20 m

5. A body of mass 3 kg moving at 4 m/s is stopped in 2 s. The average force is: 1

(a) 3 N **(b)** 12 N **(c)** 24 N **(d)** 6 N

6. A skier of mass 60 kg slides down from rest through a vertical drop of 20 m. The speed at the bottom if friction is negligible (g = 10) is: 1

(a) 20 m/s **(b)** 40 m/s **(c)** 400 m/s **(d)** 10 m/s

7. A sound wave has a frequency of 2000 Hz and a speed of 340 m/s. The number of waves in a 3.4 m length is: 1

(a) 40 **(b)** 10 **(c)** 20 **(d)** 2

8. Dividing cells of a root tip are best seen under the microscope in the: 1

(a) root cap **(b)** root hair zone **(c)** growing region **(d)** mature root

9. A stem with a girth increasing yearly has active: 1

(a) vascular cambium **(b)** stomata **(c)** root cap **(d)** apical meristem

10. A plant is described as having vascular bundles scattered in the stem, parallel venation and fibrous roots. It is a: 1

(a) pteridophyte **(b)** dicot **(c)** monocot **(d)** gymnosperm

11. An egg is fertilised on the 14th day of the menstrual cycle. The next menstruation will be: 1

(a) absent **(b)** on the 28th day **(c)** on the 21st day **(d)** on the 15th day

12. Which of these processes removes nitrogen gas from the air for plants? 1

(a) respiration **(b)** burning

(c) nitrogen fixation **(d)** transpiration

13. The predictions of Newton's law of gravitation matched the observed motion of planets. This: 1

(a) has no effect **(b)** supports the law
(c) refutes it **(d)** proves it forever

14. Which technique separates a mixture of iodine and sand? 1

(a) sublimation **(b)** magnet **(c)** filtration **(d)** decantation

15. The formula of ammonium phosphate is: 1

(a) NH_4PO_4 **(b)** $(NH_4)_3PO_4$ **(c)** $(NH_4)_2PO_4$ **(d)** $NH_4(PO_4)_3$

16. An element with atomic number 14 has valency: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 8

17. A car travels half the distance at 40 km/h and the other half at 60 km/h. The average speed is: 1

(a) 45 km/h **(b)** 48 km/h **(c)** 52 km/h **(d)** 50 km/h

18. A person weighs 600 N on the Earth (g = 10). On a planet where g is one-third of that on Earth, the mass of the person is: 1

(a) 600 kg **(b)** 20 kg **(c)** 60 kg **(d)** 180 kg

19. A body of mass 4 kg is lifted through 3 m and then carried horizontally 5 m. The total work done against gravity is (g = 10): 1

(a) 120 J **(b)** 320 J **(c)** 0 **(d)** 200 J

20. A wave of wavelength 2 m and speed 340 m/s has a time period of: 1

(a) 0.00588 s **(b)** 0.0059 Hz **(c)** 170 s **(d)** 680 s

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Calculate the time taken for sunlight to reach Earth: distance 1.5 x 10^11 m and speed 3 x 10^8 m/s, in minutes. 2

22. Why do scientists express large numbers in scientific notation? Write 300000000 in this form. 2

23. Why is the thermometer bulb placed at the side arm opening in distillation? 2

24. Why is the formula of water not HO? Explain with valency. 2

25. Calculate the average atomic mass of an element with isotopes 63 u (70 per cent) and 65 u (30 per cent). 2

26. An object is dropped from 20 m. Find the time to hit the ground and the speed on impact (g = 10). 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. An atom has 3 shells. The valence shell has 3 electrons. Find the atomic number and the valency of the element. 3

28. A car A starts from rest with a = 2 m/s squared and car B moves at uniform 20 m/s. When will A overtake B if both start together from the same point? 3

29. A truck of mass 5000 kg moving at 10 m/s collides and stops in 0.5 s. Find the average force and compare with its weight. 3

30. A lever has a load of 100 N at 20 cm from the fulcrum. Find the effort at 50 cm and at 1 m on the other side and compare MA. 3

31. Calculate the time interval between a lightning flash and the thunder if a lightning strikes 3.4 km away (v = 340 m/s). 3

OR

Explain two major differences between a bacterial cell and an animal cell with the genetic material in each. 3

32. Explain how the tissue organisation is different in a multicellular plant and an animal with examples of conducting tissues. 3

OR

An animal has a four-chambered heart, feathers and hollow bones. Identify the class and mention how the bones are adapted. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain the origin of friction, the factors on which it depends, and calculate the friction force on a 20 kg body with a coefficient of friction 0.3 (g = 10) and the acceleration if pulled with 80 N. 5

34. Explain the efficiency of a machine. A machine is supplied with 800 J and does 600 J of useful work. Find the efficiency and the energy lost. 5

35. Explain, with graphs described in words, how two sounds with different amplitudes and frequencies differ. Sketch the case for A (amplitude 2, frequency 100 Hz) and B (amplitude 1, frequency 200 Hz). 5

36. Explain how the cell is organised as a miniature factory, giving the roles of the nucleus, ribosomes, ER, Golgi, mitochondria, lysosomes and the membrane. 5

OR

Describe the adaptations of the tissues in aquatic, desert and land plants with examples. 5

37. Describe the adaptations of the birds for flight and of the fish for aquatic life with examples. 5

OR

Explain the structure of seeds (bean and maize), the parts, and the process of germination. 5

38. Calculate the amount of CO2 absorbed by a forest of 100 hectares if each hectare absorbs 6 tonnes per year; compare it with the emission of 600 cars each emitting 2.4 tonnes per year. 5

OR

Explain the use of estimation in science with three worked examples and discuss the sources of uncertainty. 5

END OF QUESTION PAPER — Total: 80 Marks

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